Module IV . Engineering Materials.

Polymers:

Combination of Indefinite number of small molecules to form a large molecule is known as **polymerization.** The large molecule is called a **polymer** and the small molecules from which polymers are formed are called **monomers.**

Polymerization are of two types.1.Addition polymerization and 2.Condensation polymerization.

1. **Addition polymerization or Chain growth polymerization**: Simple molecules containing double or triple bonds join together by breaking up of the multiple bonds to form long chain polymers is called addition polymerization. Polymerization is initiated by heat, light, pressure or a catalyst. There is no liberation of small molecules during addition polymerization. Examples are:

   a) **Polyethene (PE) :** Polymerization of ethylene at high temperature and pressure.

$$n\ CH_2{=}CH_2 \longrightarrow (CH_2-CH_2)_n-$$

By using free radical initiator, low density LDPE and ionic catalysts give high density HDPE are obtained.

Properties: PE is resistant to strong acids and alkalies at room temperature, It is a thermoplastic and can be moulded to any shape, soluble in organic solvents. Uses: PE can be used for electric insulation, polythene bags, carpets, blankets, in moulded articles etc.

c). Polyvinylchloride (PVC) formed from vinyl chloride.

$$n\ CH_2{=}CHCl \longrightarrow (CH_2-CHCl)_n-$$

Properties: It is colourless, non-inflammable, chemically inert, powder. It is resistant to light, inorganic acids, alkalies, soluble in organic solvents and brittle. Uses: It is used for making plastic pipes, rain coats, metal coatings, tank linings, safety helmets, refrigerator components, tyres, cycle and motorcycle mud guards etc.

d). Polyvinyl acetate (PVA) .Prepared by polymerization of vinyl acetate in presence of peroxides.

$$n\ CH_2{=}CH(-COOCH_3) \longrightarrow -[CH_2-CH(-COOCH_3)]_n-$$

It is a transparent , amorphous polymer, soluble in organic solvents. It is used to make water bases emulsion paints, textile fibres, leather, chewing gums etc.

2. **Condensation polymerization or Step growth polymerization.**

   Two or more polyfunctional monomers (molecules containing more than one functional group) react together to form polymers with the elimination of simple molecules like H_2O, HCl, Methanol, NH_3 etc.

a). Polyesters: These are polymers containing "ester " functional group in their main chain formed by the condensation of polyhydric alcohols with dicarboxylic acids.

<u>PET (Polyethylene terephthalate)</u>. Formed by the polymerization of ethylene glycol and Terephthalic acid.

H_2O molecule is eliminated.

$$\underset{\text{Ethylene glycol}}{n\ HO-CH_2-CH_2-OH} + \underset{\text{Terephthalic acid}}{n\ HOOC-(C_6H_4)-COOH} \longrightarrow$$

$$\underset{\text{Polyethylene terephthalate (PET)}}{-(O-CH_2-CH_2-OOC-(C_6H_4)-CO-)_n-}$$

<u>Properties</u>: good fibre forming material, highly resistant to mineral and organic acids,but less resistant to alkalies and have high stretch resistance. <u>Uses</u>: Polyester threads are used in dress materials and home furnishings, PET bottles and containers, films, tarpaulin, film insulation and so on. Blendable with cotton and wool which enhances the life of cotton and woolen clothes.

<u>Nylon</u> is used as a general name for all fiber forming polyamides having protein-like structure.

<u>Nylon-6,6.</u> Is formed from Adipic acid and hexamethylene diamine in the presence of catalyst.

$$\underset{\text{Adipic acid}}{n\ HOOC-(CH_2)_4-COOH} + \underset{\text{hexamethylene diamine}}{n\ NH_2-(CH_2)_6-NH_2} \longrightarrow$$

$$\underset{\text{Nylon-6,6.}}{-(OC-(CH_2)_4-CONH-(CH_2)_6-NH)_n-}$$

Properties of Nylons: Nylons are light polymers, insoluble in common solvents, soluble in phenol and formic acid, absorbs little moisture, very flexible and can be blended with wool.

Uses: Nylon-6,6. Is used as fibers for making garments, socks, carpets etc. Nylon-6 is used for making ropes, nylon bearings, tyre-cords, films etc.

<u>Mechamism of polymerization.</u>

Four mechanisms are proposed for polymerization of ethylene and substituted compounds. Polymerization take place through three principal stages.

1).Chain Initiation step

2).Chain propagation step and

3. Chain termination step.

1. **Free radical mechanism**: Polymerization by catalysts such as benzoyl peroxide,H_2O_2 initiates free radical mechanism. These catalysts are called free radical generators.

<u>1).Chain Initiation step</u>: Two reactions are involved in chain initiation step.

a). Production of free radicals by the homolytic fission of catalyst (chain initiator).

$R\text{-}CO.O\text{-}O.OC\text{-}R \rightarrow 2R^0 + 2CO_2$

b). The addition of free radical to first monomer to produce chain initiator species.

$R^0 + CH_2{=}CH_2 \rightarrow R-CH_2-CH_2^0$ ....(chain initiator)

2). Chain propagation step: Chain growth by successive addition of large numbers of monomer molecules.

$R-CH_2-CH_2^0 + n\,CH_2{=}CH_2 \rightarrow R-(CH_2-CH_2)_n-CH_2-CH_2^0$

3).Chain Termination step: It is difficult to control the chain length. The following chain process expected.

$R^0 + R^0 \rightarrow R-R$

$R-CH_2-CH_2^0 + R^0 \rightarrow R-CH_2-CH_2-R$

$R-CH_2-CH_2^0 + R-CH_2-CH_2^0 \rightarrow R-CH_2-CH_2-CH_2-CH_2-R$

$R-CH_2-CH_2^0 + R-(CH_2-CH_2)_n-CH_2-CH_2^0 \rightarrow$

$R-(CH_2-CH_2)_n-CH_2-CH_2-CH_2-CH_2-R$

$R-(CH_2-CH_2)_n-CH_2-CH_2^0 + R^0 \rightarrow R-(CH_2-CH_2)_n-CH_2-CH_2-R$

$R-(CH_2-CH_2)_n-CH_2-CH_2^0 + R-(CH_2-CH_2)_n-CH_2-CH_2^0 \rightarrow$

$R-(CH_2-CH_2)_n-CH_2-CH_2-(CH_2-CH_2)_n-CH_2-CH_2-R$

2. **Ionic mechanism:** Polymerization is brought about by catalysts which produce ions for chain propagation. Depending upon the ions produced , there are two types of ionic mechanisms.

a) Anionic mechanism b). Cationic mechanism

a) **Anionic mechanism:** When monomer contains electron withdrawing groups such as $-Cl, -COOH, -CO, -COOMe$, polymerization take place by using anionic initiators(catalysts) like alkyl lithium, Grignard reagent etc. Polar solvents increase the rate of polymerization.

1).Chain Initiation step: Two reactions are involved in chain initiation step. Carbanion is formed as chain initiator.

a). $R-Li \rightarrow R^{(-)} + Li^+$

b). $R^{(-)} + CH_2{=}CH-X \rightarrow R-CH_2-CH^{(-)}-X$ (where X is an electron withdrawing group)

2).Chain propagation step:

$R-CH_2-CH^{(-)}-X + n\,CH_2{=}CH-X \rightarrow R-(CH_2-CH)_n-CH_2-CH^{(-)}-X$

3). Chain termination step: Water or alcohol is added to terminate the polymerization.

$R-(CH_2-CH)_n-CH_2-CH^{(-)}-X \xrightarrow{H^+} R-(CH_2-CH)_n-CH_2-CH_2-X$

b).Cationic mechanism: When monomer contains electron donating groups such ether group or benzene ring polymerization take place by using cationic initiators like H_2SO_4, HF, Lewis acids like $AlCl_3$, BF_3 etc.

1).Chain Initiation step: Polymerization proceeds via carbocation as intermediate as follows.

a). Formation of cationic initiator Y^+ from catalyst. $BF_3 + R\!-\!OH \rightarrow R^+ + BF_3OH$

b). $R^+ + CH_2{=}CH\!-\!Z \rightarrow R\!-\!CH_2\!-\!CH^+\!-\!Z$ (where Z is an electron donating group)

2).Chain propagation step

$$R\!-\!CH_2\!-\!CH^+\!-\!Z + n\,CH_2{=}CH\!-\!Z \rightarrow R\!-\!(CH_2\!-\!CH\!-\!Z)_n\!-\!CH_2\!-\!CH^+\!-\!Z$$

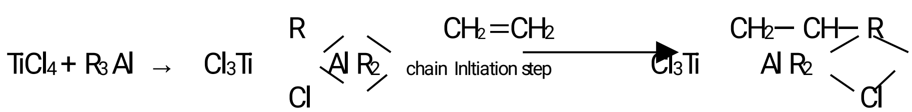
3). Chain termination step: Termination take place by the loss of a proton resulting in the formation of a double bond at the end of the chain.

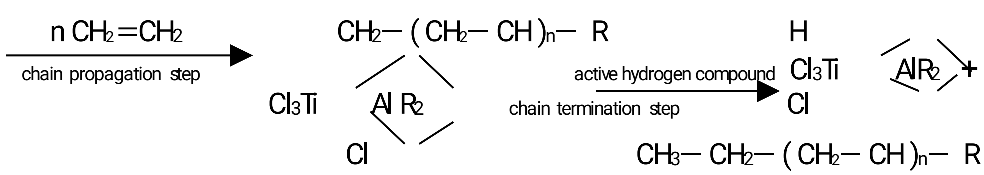
$$R\!-\!(CH_2\!-\!CH\!-\!Z)_n\!-\!CH_2\!-\!CH^+\!-\!Z \rightarrow R\!-\!(CH_2\!-\!CH)_n\!-\!CH_2{=}CH\!-\!Z$$

3. **Coordination mechanism of polymerization (Ziegler-Natta Polymerization).**

   Titanium tetrachloride $TiCl_4$ and trialkyl aluminium R_3Al are together known as Ziegler Natta catalyst.

The mechanism is as follows: (1) Chain Initiation step:





Chain termination takes place with an active hydrogen compound.

Classification of Polymers

There are many types of classification.

1. Based on source : Natural and Synthetic Polymers.
2. Based upon synthesis : Addition polymers and condensation polymers.
3. Based upon elements : Inorganic and Organic polymers.
4. Based upon building units : Homo polymers and Copolymers.
5. Based upon thermal behaviour: (a) Plastics (b) Fibers (c). Elastomers.

Difference between Addition polymers and Condensation polymers.

	Addition polymers	Condensation polymers
1.	Formed from monomers containing	1.Formed from monomers containing

	double or triple bonds.	poly functional groups.
2.	They are homo polymers.	2.They are copolymers of at least two different types of monomers.
3.	No elimination of bye-products like water or NH_3 etc	3.Always simple molecules like H_2O, NH_3, HCl etc are eliminated during polymerization
4.	They are long chain thermoplastics	4.They are long chain or cross linked three dimensional polymers.
5.	Ex: Polyethene, PVC, PVA etc.	Ex: Nylons, Polyesters, Bakelite etc.

Plastics are of two types. Thermoplastics and thermosetting plastics

	Thermoplastics	Thermosetting plastics
1	Formed by addition polymerization.	1.Formedbycondensation polymerization.
2	Have long chain linear structure.	2.Three dimensional cross linking structure.
3	They are soft, weak and less brittle.	3.They are hard,strong and brittle.
4	Can be remoulded, recast and reused by suitable process	4.Can not be remoulded or reused.
5	Soluble in organic solvents.	5.Insoluble in organic solvents.
6	Ex: PVC, PVA, Polyethene etc.	6.Ex: Nylons, Polyesters, Bakelite etc.

Natural Polymer- Rubber

Natural rubber is an elastomer obtained as latex from the bark of rubber trees. The milky white latex is collected in a pan and coagulated by agitating with formic acid or acetic acid. The coagulated rubber is squeezed by means of rollers, washed, dried and smoked to prevent fungal attack.

Natural rubber is polyisoprene. Its molecular weight is of the order 300000.

$$\underset{\text{Isoprene}}{CH_2=C(CH_3)-CH=CH_2} \longrightarrow \underset{\text{Rubber}}{(CH_2-C(CH_3)=CH-CH_2)_n-}$$

Rubber is elastic due to the coiled form of rubber chain. This straightens on stretching and on releasing the chain reverts to original coiled form.

Properties: Natural rubber is soft, sticky and insoluble in water, alcohol, dil. acids and alkalies, soluble in CS_2, CCl_4, petrol and turpentine. It has low tensile strength.

Drawbacks of raw rubber: (1) It is plastic in nature. (2).It becomes soft at high temperature and brittle at low temperature. So it can be used in 10 to 60°C. (3). It has large water absorption capacity.(4).Oxidized by HNO_3, Con.H_2SO_4.(5). It degrades on exposure to air. (6). It swells in organic solvents and degrades. (7). It possesses tackiness and less durability.

Uses: Natural rubber is used to make erasers, elastic threads, balloons, rubber bands etc.

Vulcanization of rubber: Rubber is heated with 5 to 8% sulphur, 5% ZnO as filler, and 0.5% nitrogen containing compound as accelerator at 400 to 440K for half an hour. By vulcanization, double bonds in rubber are cross linked with sulphur between chains. Vulcanized rubber has better resistance to moisture, oxidation and low elasticity.

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\begin{array}{l}
-(CH_2-C(CH_3)=CH-CH_2-CH_2-C(CH_3)=CH-CH_2)_n- \\
\qquad\qquad\qquad + \\
-(CH_2-C(CH_3)=CH-CH_2-CH_2-C(CH_3)=CH-CH_2)_n- \\
\qquad \text{raw rubber}
\end{array}
\xrightarrow[5\%\,S,\ 5\%ZnO]{440K}
$$

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\left\{
\begin{array}{cccccccc}
-CH_2- & C(CH_3)- & CH- & CH_2- & CH_2- & C(CH_3)- & CH- & CH_2- \\
 & | & | & & & | & | & \\
S & & S & & S & & S & \\
| & & | & & | & & | & \\
-CH_2- & C(CH_3)- & CH- & CH_2- & CH_2- & C(CH_3)- & CH- & CH_2-
\end{array}
\right\}_n
$$

vulcanized rubber

Advantage of vulcanization:1). Prevent slippage (2). Make it less sensitive to temperature. useful in the range -40^o to 100^oC. (3) has only slight tackiness,(4)low elasticity(5). Less water absorption capacity (6). resistant to organic solvents (7) High resistance to wear and tear.

(8) recovers original shape after the deforming load is removed.

(9).easy to produce articles of desired shapes. (10). Better electric insulator.

Polymer Processing methods

Conversion of a polymer into various commercial products is called polymer processing. Polymer processing consists of three stages.

I. The plasticization stage

The solid polymer in the form of powder or granules is uniformly melted without decomposing the polymer, additives are added and mixed thoroughly. This process is called blending or compounding.

<u>Compounding of plastics</u>

The process of mixing the plastics with various other substances to improve the workability, flexibility, and to make it colourful, is called compounding. The main types of compounding ingredients and their functions are given below.

1. **Catalysts or accelerators**- are added to accelerate the polymerization usually for themosets. Usual catalysts are H_2O_2, benzoyl peroxide, acetyl sulphuric acid, ZnO, metals like Na, Ag, Cu, Pb etc.
2. **Plasticizers** - are added to increase the plasticity and flexibility. The most commonly used plasticizers are vegetable oils , organophosphates, esters etc.
3. **Fillers** – are added to give better hardness, tensile strength, workability and finish. Some examples are Asbestos powder, carborundum, quartz, mica, wood flour, china clay, gypsum, marble powder, metal powder, cotton threads, scraps of clothes etc
4. Lubricants - like waxes, oils and soaps are added to impart glossy finish and antisticking property to plastics.
5. Colouring materials such as organic dyes and inorganic pigments are added to get a better colour to plastics.
6. Stabilizers- are added to improve the thermal stability during moulding process. Lead, leadchromate, red lead, stearates of lead. Cadmium, barium etc are used as stabilizers.
7. Resin- is a binder which holds the different constituents together

II.Casting stage : The molten liquid polymer is moulded into desired shape.

III.The cooling stage : cooling is done by spraying water or by circulating air by which the liquid sets into desired shape.

II. Casting Or Moulding Techniques

1. **Extrusion moulding:** This method is used for the production of objects of uniform cross section like pipes, tubes, fibers, electric cables etc

   The molten polymer liquid is forced into the die under pressure by means of a metal screw conveyor inside a barrel. Allowed to cool and set after moving from the die and cut into suitable length.

2 **Compression moulding:** This method is used to mould large parts which are flat or smoothly curved articles Like fiber glass body parts for automobiles, boats, electrical switch panels etc.

Large sheets of thermoplastic polymer material are pre-heated and placed over one half of the mould and the other half of the mould is pressed over it so that the soft polymer material spreads and take up the desired shape of the mould. After moulding and cooling article is taken out by opening the mould parts. It is a low cost method where the wastage is minimum and can be applied to both thermoplastic and thermosetting resins

3.**Injection moulding:** This method is used to manufacture all types of plastic products with complicated shapes but one piece at a time. The molten polymer from the heating cylinder is injected at high pressure at a controlled rate through a small hole using a piston plunger or a screw arrangement into the suitable mould made of steel or aluminium. It is then allowed to cool . The mould can be opened up to take the finished product out. This method is widely used because of high speed, low mould cost, low loss of material and low finishing cost. Plates, mugs, buckets, stools, chairs etc are moulded by this method.

4.**Transfer moulding:** Pre weighed amount of polymer powder is heated in a separate chamber called transfer pot and then forced into a pre heated mould through an orifice into the mould cavity by a plunger at high pressure .It cures and sets in the cavity and then ejected out. This method is mainly used for moulding articles made of thermosetting plastics.

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3 **Blow moulding:** used for moulding bottles of various shapes. A die similar to bottle is used. The polymer is extruded into the two halves of an open blow mould of the bottle. When the extrusion reaches proper length , the mould closes holding the neck end opened and the bottom end closed. A rod –like pin is inserted into the neck of the hot mould to form a threaded opening. After the bottle cools, the mould opens to eject the bottle out. Excess plastic is trimmed.